

PART E — Data Format (CSV Schema)

The CSV file contains **15 columns**.

Column 1: t_ms

Column 2: imu1_acc_x

Column 3: imu1_acc_y

Column 4: imu1_acc_z

Column 5: imu1_gyr_x

Column 6: imu1_gyr_y

Column 7: imu1_gyr_z

Column 8: imu2_acc_x

Column 9: imu2_acc_y

Column 10: imu2_acc_z

Column 11: imu2_gyr_x

Column 12: imu2_gyr_y

Column 13: imu2_gyr_z

Column 14: emg_env

Column 15: temp_c

The first row will be the header:

t_ms,imu1_acc_x,imu1_acc_y,imu1_acc_z,imu1_gyr_x,imu1_gyr_y,imu1_gyr_z,imu2_acc_x,imu2_acc_y,imu2_acc_z,imu2_gyr_x,imu2_gyr_y,imu2_gyr_z,emg_env,temp_c

Each row after that represents one time sample.

Units:

t_ms = milliseconds since ESP32 boot

imu*acc* = accelerometer raw output (mg scale from SparkFun library)

`imu*gyr*` = gyroscope raw output (deg/s from SparkFun library)

`emg_env` = MyoWare envelope output from ADC (0–4095)

`temp_c` = temperature in degrees Celsius from TMP117

Sampling rate is approximately 50 Hz (one row every ~20 ms).